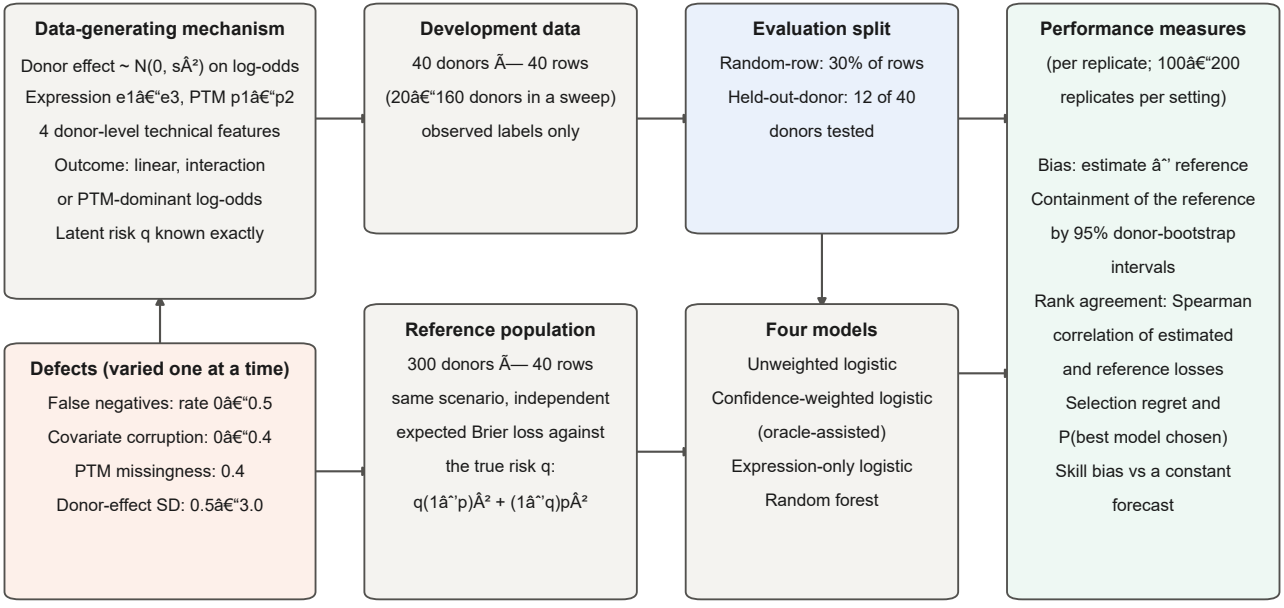
Data-generating mechanism
Donor effect ~ N(0, sÂ²) on log-odds
Expression e1â€“e3, PTM p1â€“p2
4 donor-level technical features
Outcome: linear, interaction
or PTM-dominant log-odds
Latent risk q known exactly
Development data
40 donors Ã— 40 rows
(20â€“160 donors in a sweep)
observed labels only
Evaluation split
Random-row: 30% of rows
Held-out-donor: 12 of 40
donors tested
Performance measures
(per replicate; 100â€“200
replicates per setting)
Bias: estimate â^' reference
Containment of the reference
by 95% donor-bootstrap
intervals
Rank agreement: Spearman
correlation of estimated
and reference losses
Selection regret and
P(best model chosen)
Skill bias vs a constant
forecast
Defects (varied one at a time)
False negatives: rate 0â€“0.5
Covariate corruption: 0â€“0.4
PTM missingness: 0.4
Donor-effect SD: 0.5â€“3.0
Reference population
300 donors Ã— 40 rows
same scenario, independent
expected Brier loss against
the true risk q:
q(1â^'p)Â² + (1â^'q)pÂ²
Four models
Unweighted logistic
Confidence-weighted logistic
(oracle-assisted)
Expression-only logistic
Random forest